APEX RETAIL GROUP | INDEPENDENT HOMEWORK ASSIGNMENT

DOCUMENT: CLASS HOMEWORK (SESSION 2)

SCENARIO: CUSTOMER SUPPORT TICKET SLA AUDIT & CORRUPTED LOG SANITIZATION

ASSIGNMENT OVERVIEW: You are assisting the Customer Service Director at Apex Retail Group. The customer support team handles inquiries categorized as either "Urgent" or "Routine".

Corporate SLAs specify:

- "Urgent" support tickets must be resolved within 4.0 hours.
- "Routine" support tickets must be resolved within 24.0 hours.

The support platform exported a raw batch log containing ticket IDs, ticket types, and resolution times (in hours). However, the export file contains data corruption due to a database synchronization error.

RAW TICKET DATASET:

```Python
ticket_batch = [
    {"ticket_id": "TCK-501", "type": "Urgent",  "resolution_hours": 2.5},
    {"ticket_id": "TCK-502", "type": "Routine", "resolution_hours": 18.0},
    {"ticket_id": "TCK-503", "type": "Urgent",  "resolution_hours": 6.0},
    {"ticket_id": "TCK-504", "type": "Routine", "resolution_hours":
"TIMEOUT_ERROR"},
    {"ticket_id": "TCK-505", "type": "Urgent",  "resolution_hours": 1.0},
    {"ticket_id": "TCK-506", "type": "Routine", "resolution_hours": 28.5},
    {"ticket_id": "TCK-507", "type": "Urgent",  "resolution_hours":
"NULL_VAL"},
    {"ticket_id": "TCK-508", "type": "Routine", "resolution_hours": 12.0},
    {"ticket_id": "TCK-509", "type": "Urgent",  "resolution_hours": 5.5},
    {"ticket_id": "TCK-510", "type": "Routine", "resolution_hours": 22.0}
]
```

PROBLEM 1: FUNCTION CREATION WITH MULTI-LEVEL LOGIC & ERROR HANDLING

In a Jupyter Notebook named `homework_session2_yourname.ipynb`:

Write a Python function `audit_ticket_sla(ticket_type, raw_hours)` that:

1. Uses `try` / `except` to attempt converting `raw_hours` to a `float`. If conversion fails, return `"Corrupted Data"`.

2. Validates if `ticket_type` is valid (must be `"Urgent"` or `"Routine"`). If not, return `"Invalid Type"`.
3. Evaluates SLA compliance:
    - For `"Urgent"` tickets: if hours > 4.0, return `"SLA Violation"`, else return `"SLA Met"`.
    - For `"Routine"` tickets: if hours > 24.0, return `"SLA Violation"`, else return `"SLA Met"`.

PROBLEM 2: AUTOMATED BATCH AUDIT PIPELINE

Write Python code using a `for` loop to process `ticket_batch`:

1. Iterate over every ticket dictionary in `ticket_batch`.
2. Pass ticket attributes to `audit_ticket_sla()`.
3. Maintain tallies for:
    - Total tickets evaluated
    - SLA Met count
    - SLA Violation count
    - Corrupted/Invalid log count
4. Print a clean, formatted audit log showing each individual ticket ID and its evaluated status.
5. Calculate and print the overall SLA Success Rate (%) across valid tickets.

PROBLEM 3: EXECUTIVE BRIEFING MEMO

Write a 150-word formal memo to the Customer Service Director containing:

1. The overall SLA Success Rate across valid customer support tickets.
2. A breakdown comparing Urgent ticket compliance versus Routine ticket compliance.
3. An explicit operational risk warning regarding system data quality and corrupted ticket logs.

SUBMISSION DEADLINE: Prior to Session 3.